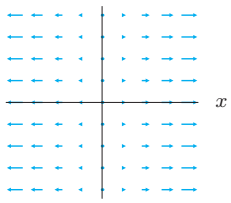
Exercises and Problems for Section 17.3

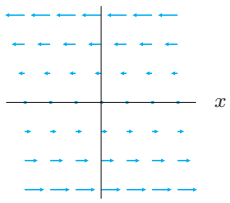
Exercises

For Exercises 1–6, find formulas for the vector fields. (There are many possible answers.)

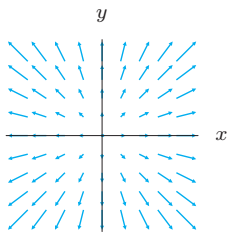
1.



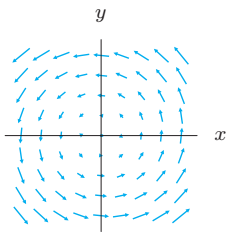
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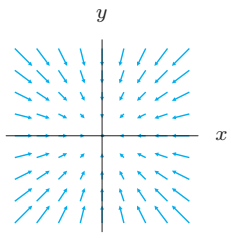
3.



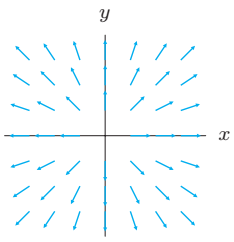
4.



5.



6.



In Exercises 7–10, assume $x, y > 0$ and decide if

(a) The vector field is parallel to the x-axis, parallel to the y-axis, or neither.

(b) As x increases, the length increases, decreases, or neither.

(c) As y increases, the length increases, decreases, or neither.

Assume $x, y > 0$.

7. $\vec{F} = x\vec{j}$

8. $\vec{F} = y\vec{i} + \vec{j}$

9. $\vec{F} = (x + e^{1-y})\vec{i}$

10. $\text{grad}(x^4 + e^{3y})$

Sketch the vector fields in Exercises 11–19 in the xy-plane.

11. $\vec{F}(x, y) = 2\vec{i} + 3\vec{j}$

12. $\vec{F}(x, y) = y\vec{i}$

13. $\vec{F}(x, y) = -y\vec{j}$

14. $\vec{F}(\vec{r}) = 2\vec{r}$

15. $\vec{F}(\vec{r}) = \vec{r}/\|\vec{r}\|$

16. $\vec{F}(\vec{r}) = -\vec{r}/\|\vec{r}\|^3$

17. $\vec{F} = y\vec{i} - x\vec{j}$

18. $\vec{F}(x, y) = 2x\vec{i} + x\vec{j}$

19. $\vec{F}(x, y) = (x + y)\vec{i} + (x - y)\vec{j}$

20. For each description of a vector field in (a)-(d), choose one or more of the vector fields I-IX.

(a) Pointing radially outward, increasing in length away from the origin.

(b) Pointing in a circular direction around the origin, remaining the same length.

(c) Pointing towards the origin, increasing in length farther from the origin.

(d) Pointing clockwise around the origin.

I. $\dfrac{x\vec{i} + y\vec{j}}{\sqrt{x^2 + y^2}}$ II. $\dfrac{-y\vec{i} + x\vec{j}}{\sqrt{x^2 + y^2}}$ III. $\vec{r}$

IV. $-\vec{r}$ V. $-y\vec{i} + x\vec{j}$ VI. $y\vec{i} - x\vec{j}$

VII. $y\vec{i} + x\vec{j}$ VIII. $\dfrac{\vec{r}}{\|\vec{r}\|^3}$ IX. $-\dfrac{\vec{r}}{\|\vec{r}\|^3}$

21. Each vector field in Figures (I)–(IV) represents the force on a particle at different points in space as a result of another particle at the origin. Match up the vector fields with the descriptions below.

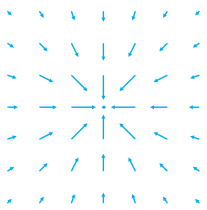
(a) A repulsive force whose magnitude decreases as distance increases, such as between electric charges of the same sign.

(b) A repulsive force whose magnitude increases as distance increases.

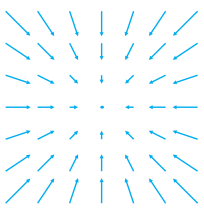
(c) An attractive force whose magnitude decreases as distance increases, such as gravity.

(d) An attractive force whose magnitude increases as distance increases.

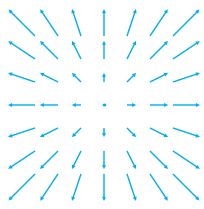
(I)



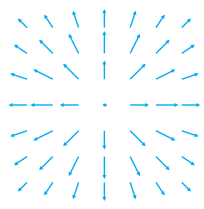
(II)



(III)



(IV)



Problems

In Problems 22–26, give an example of a vector field $\vec{F}(x, y)$ in 2-space with the stated properties.

22. $\vec{F}$ is constant

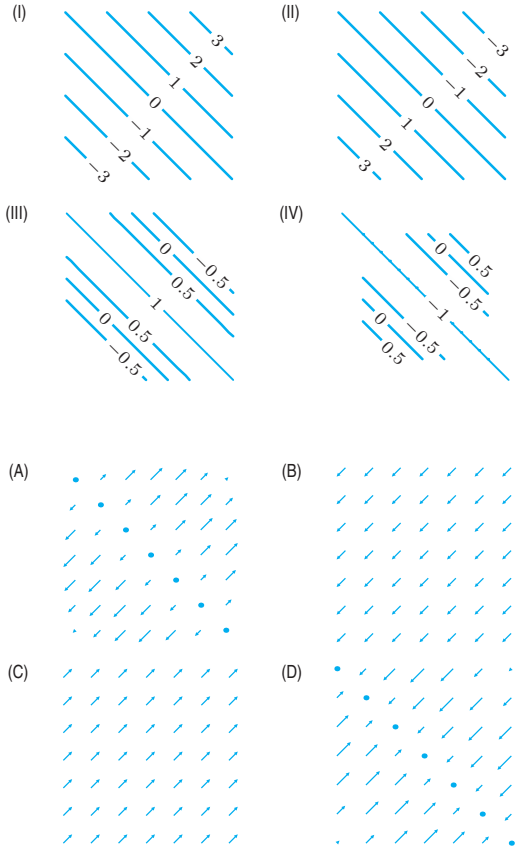
23. $\vec{F}$ has a constant direction but $\|\vec{F}\|$ is not constant

24. $\|\vec{F}\|$ is constant but $\vec{F}$ is not constant

25. Neither $\|\vec{F}\|$ nor the direction of $\vec{F}$ is constant

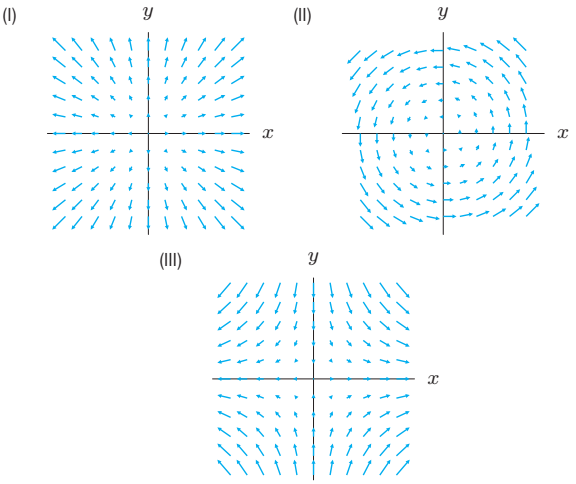
26. $\vec{F}$ is perpendicular to $\vec{G} = (x + y)\vec{i} + (1 + y^2)\vec{j}$ at every point

27. Match the level curves in (I)–(IV) with the gradient fields in (A)–(D). All figures use the same square window.



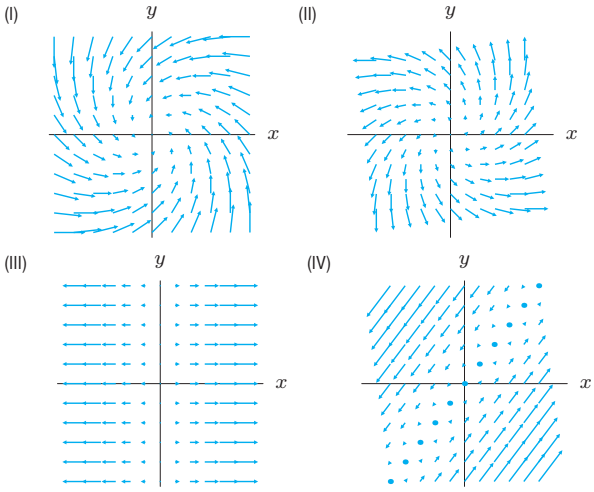
Problems 28–29 concern the vector fields $\vec{F} = x\vec{i} + y\vec{j}$, $\vec{G} = -y\vec{i} + x\vec{j}$, and $\vec{H} = x\vec{i} - y\vec{j}$.

28. Match $\vec{F}$, $\vec{G}$, $\vec{H}$ with their sketches in (I)–(III).



29. Match the vector fields with their sketches, (I)–(IV).

(a) $\vec{F} + \vec{G}$ **(b)** $\vec{F} + \vec{H}$ **(c)** $\vec{G} + \vec{H}$ **(d)** $-\vec{F} + \vec{G}$



In Problems 30–32, write formulas for vector fields with the given properties.

30. All vectors are parallel to the x-axis; all vectors on a vertical line have the same magnitude.

31. All vectors point toward the origin and have constant length.

32. All vectors are of unit length and perpendicular to the position vector at that point.

33. **(a)** Let $\vec{F} = x\vec{i} + (x + y)\vec{j} + (x - y + z)\vec{k}$. Find a point at which $\vec{F}$ is parallel to l, the line $x = 5 + t$, $y = 6 - 2t$, $z = 7 - 3t$.

(b) Find a point at which $\vec{F}$ and l are perpendicular.
(c) Give an equation for and describe in words the set of all points at which $\vec{F}$ and l are perpendicular.

In Problems 34–35, let $\vec{F} = x\vec{i} + y\vec{j}$ and $\vec{G} = -y\vec{i} + x\vec{j}$.

34. Sketch the vector field $\vec{L} = a\vec{F} + \vec{G}$ if:

(a) $a = 0$ **(b)** $a > 0$ **(c)** $a < 0$

35. Sketch the vector field $\vec{L} = \vec{F} + b\vec{G}$ if:

(a) $b = 0$ **(b)** $b > 0$ **(c)** $b < 0$

36. In the middle of a wide, steadily flowing river there is a fountain that spouts water horizontally in all directions. The river flows in the $\vec{i}$-direction in the xy-plane and the fountain is at the origin.

(a) If $A > 0$, $K > 0$, explain why the following expression could represent the velocity field for the combined flow of the river and the fountain:

$$\vec{v} = A\vec{i} + K(x^2 + y^2)^{-1}(x\vec{i} + y\vec{j}).$$

(b) What is the significance of the constants A and K?
(c) Using a computer, sketch the vector field $\vec{v}$ for $K = 1$ and $A = 1$ and $A = 2$, and for $A = 0.2$, $K = 2$.

37. Figures 17.32 and 17.33 show the gradient of the functions $z = f(x, y)$ and $z = g(x, y)$.

(a) For each function, draw a rough sketch of the level curves, showing possible z-values.
(b) The xz-plane cuts each of the surfaces $z = f(x, y)$ and $z = g(x, y)$ in a curve. Sketch each of these curves, making clear how they are similar and how they are different from one another.

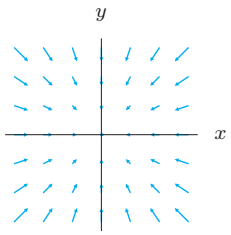


Figure 17.32: Gradient of $z = f(x, y)$

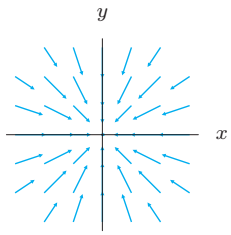


Figure 17.33: Gradient of $z = g(x, y)$

38. Let $\vec{F} = u\vec{i} + v\vec{j}$ be a vector field in 2-space with magnitude $F = \|\vec{F}\|$.

(a) Let $\vec{T} = (1/F)\vec{F}$. Show that $\vec{T}$ is the unit vector in the direction of $\vec{F}$. See Figure 17.34.
(b) Let $\vec{N} = (1/F)(\vec{k} \times \vec{F}) = (1/F)(-v\vec{i} + u\vec{j})$. Show that $\vec{N}$ is the unit vector pointing to the left of and at right angles to $\vec{F}$. See Figure 17.34.

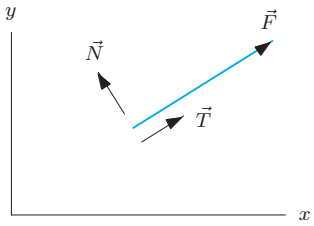


Figure 17.34

Strengthen Your Understanding

In Problems 39–40, explain what is wrong with the statement.

39. A plot of the vector field $\vec{G}(x, y, z) = \vec{F}(2x, 2y, 2z)$ can be obtained from a plot of the vector field $\vec{F}(x, y, z)$ by doubling the lengths of all the arrows.

40. A vector field $\vec{F}$ is defined by the formula $\vec{F}(x, y, z) = x^2 - yz$.

In Problems 41–42, give an example of:

41. A nonconstant vector field that is parallel to $\vec{i} + \vec{j} + \vec{k}$ at every point.

42. A nonconstant vector field with magnitude 1 at every point.

17.4 THE FLOW OF A VECTOR FIELD

When an iceberg is spotted in the North Atlantic, it is important to be able to predict where the iceberg is likely to be a day or a week later. To do this, one needs to know the velocity vector field of the ocean currents, that is, how fast and in what direction the water is moving at each point.

In this section we use differential equations to find the path of an object in a fluid flow. This path is called a flow line. Figure 17.35 shows several flow lines for the Gulf Stream velocity vector field in Figure 17.18 on page 937. The arrows on each flow line indicate the direction of flow.